Exercise Sheet

Task 1 – Hamming Codes

a) Use the table to encode these bitstrings: $(10101110001)_2$, $(11011010011)_2$

> $(10101110001)_2$: $(101011110000101)_2$
> $(11011010011)_2$: $(110110110010101)_2$

b) Use the table to decide at which position an error occurred: $(101101101010101)_2$, $(111010110101011)_2$

Bit: position	8	4	2	1	Bit: value
3	X	X			
5	X		X		
6	X			X	
7	X				
9		X	X		
10		X		X	
11		X			
12			X	X	
13			X		
14				X	
15					
Parity bits (computed):					
Parity bits (recieved):					

> $(101101101010101)_2$: 13
> $(111010110101011)_2$: 7

Task 2 – Bit-PLRU

Have a look at the following sequence of accesses:

0xAB47, 0xB759, 0xC7D5, 0xB759, 0xD8AA, 0xEBFF, 0xC7D5, 0xFFF1, 0xC7D5, 0xD8AA, 0xEBFF, 0xAB47, 0x7A39, 0xC7D5

a) Fill in the Cache table for a fully associative cache that has 4 cache lines and uses Bit-PLRU as replacement policy.
Decide how many misses occur and of what type these misses are.

b) What is the minimum number of misses for this access sequence in a cache with 4 lines? (Hint: Optimal Replacement Policy)

Access	Line 1	Status	Line 2	Status	Line 3	Status	Line 4	Status	Hit/Miss(-type)
0xAB47									
0xB759									
0xC7D5									
0xB759									
0xD8AA									
0xEBFF									
0xC7D5									
0xFFF1									
0xC7D5									
0xD8AA									
0xEBFF									
0xAB47									
0x7A39									
0xC7D5									

Access	Line 1	Status	Line 2	Status	Line 3	Status	Line 4	Status	Hit/Miss(-type)
0xAB47	0xAB47	1		0		0		0	Compulsory
0xB759		1	0xB759	1		0		0	Compulsory
0xC7D5		1		1	0xC7D5	1		0	Compulsory
0xB759		1		1		1		0	Hit
0xD8AA		0		0		0	0xD8AA	1	Compulsory
0xEBFF	0xEBFF	1		0		0		1	Compulsory
0xC7D5		1		0		1		1	Hit
0xFFF1		0	0xFFF1	1		0		0	Compulsory
0xC7D5		0		1		1		0	Hit
0xD8AA		0		1		1		1	Hit
0xEBFF	0xEBFF	1		0		0		0	Hit
0xAB47		1	0xAB47	1		0		0	Capacity
0x7A39		1		1	0x7A39	1		0	Compulsory
0xC7D5		0		0		0	0xC7D5	1	Capacity

9 Misses occur when using Bit-PLRU. The minimum number of misses possible for this sequence is 8, which can be determined by mapping this sequence to a cache with 4 lines that uses the optimal replacement policy.